# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Reviewed URS, FRS, and design specifications for a **GxP-regulated life sciences system**, identifying **15+ gaps** in requirements and testability prior to formal execution.
- Authored **12+ validation test protocols**, developing positive and negative test scenarios based on system risk and requirement criticality, ensuring adequate traceability.
- Executed formal qualification testing in **controlled QA environments**, independently assessing results against acceptance criteria and documenting outcomes with minimal supervision.
- Performed protocol dry runs on lower environments to verify accuracy and completeness, identifying **7 requirement gaps** and refining test coverage before formal testing.
- Managed **20+ deviations and defects** through their lifecycle, including root cause analysis, retesting, and final closure, ensuring compliance with GxP standards.
- Collaborated with **Business Analysts and System Owners** to clarify functional requirements and ensure testability for enterprise systems.
- Containerized validation tooling using **Docker** for consistent deployment across development and QA environments.

## SKILLS

**Validation & Testing** – Computer System Validation (CSV), Computer Software Assurance (CSA), GxP, Test Protocol Authoring, Test Execution, Requirements Analysis, Risk-Based Validation, Dry Runs, Deviation Management, Defect Tracking, UAT Support, GAMP 5, EU GMP Annex 11, FDA 21 CFR Part 11, Data Integrity

**Software Development** – Java, Spring Boot, REST APIs, SQL, PostgreSQL, Python, Docker, Git, Jenkins

**Cloud & Infrastructure** – AWS (EC2, S3, Lambda, CloudWatch), Linux, Shell Scripting

**Data Analysis & BI** – Power BI, Tableau, DAX, Power Query, R/RStudio, pandas, NumPy, scikit-learn

**Enterprise Systems** – ERP, MES, LIMS, QMS

## PROJECTS

### GxP System Validation Protocol Development (HCL Technologies)

- Developed comprehensive validation test protocols for a **financial-services operations dashboard**, focusing on transaction monitoring and reconciliation workflows.
- Authored detailed test cases aligned with GxP principles, ensuring traceability from URS to FRS for critical system functions.
- Collaborated with business analysts to refine acceptance criteria and ensure testability for operational reporting features.

### Automated Test Data Generation for ERP Integration (HCL Technologies)

- Created **Python scripts** to generate realistic test data for validating integrations between an ERP system and other enterprise applications.
- Implemented data generation logic to cover various edge cases and business scenarios, improving test coverage for cross-system data flows.
- Utilized **SQL** to populate and validate test databases, ensuring data integrity and consistency across integrated platforms.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)